Chapter 7 Troubleshooting

This chapter provides a simple introduction to avoiding and handling such problems. More detailed information on troubleshooting particular facilities is provided in the chapters describing how to use them. For information on troubleshooting operation from MS-DOS, see the *MS-DOS Programmer's Manual*.

7.1 Avoiding Problems

Hunter 16 is designed, built and quality tested for robustness in every sense including robustness of program execution. Its physical construction protects it against mechanical disturbance, while the conventional computer's Achilles heel - external electrical interference is virtually eliminated.

Hunter 16 has been designed and tested to provide a powerful, yet simple-to-operate package. However, as with any complex system, things may go wrong, particularly while you are unfamiliar with its operation.

The best way to deal with problems is to avoid them!

Follow these few simple rules and you should find that life is relatively trouble-free:

- Treat **Hunter 16** with respect use it in the recommended operating conditions, don't drop it or drop anything on it!
- Observe the recommendations on charging and use made in section 6.5.
- Read the operating instructions for software and peripherals before using them.
- When expanding your system, always use compatible software of good quality.

Once you start using **Hunter 16** for serious work, you should also take out insurance' against your own mistakes, software or hardware failure, by making copies or backups of important data at regular intervals. These copies would usually take the form of files stored on external disks or host computers, but printed listings or even handwritten notes may be sufficient if the amount of data is small.

There are some specific situations that can cause **Hunter 16** to mis-execute an application, or 'crash'. The commonest causes are as follows.

7.1.1 Programming errors

Applications may include bugs (such as invalid jump instructions,) that would crash any computer. Programmers are of course responsible for the logical correctness of their programs.

Applications may make invalid system calls, with unpredictable results. Programmers must remain within the programming guidelines included in this guide and the *MS-DOS Programmer's Manual* if they wish their programs to behave correctly.

7.1.2 Physical abuse

Ingress of water, corrosion of internal parts, damage to components through excessive shock or persistent high level vibration can reduce electronic tolerances.

7.1.3 Operation Outside Specified Temperature Range

Hunter 16 is specified for operation in the range 0 to 55 degrees Centigrade.

Operation at higher temperatures reduces tolerances, while operation a lower temperatures severely restricts battery capacity and slows LCD screen response time substantially. **Hunter 16** is however unlikely to mis-execute program simply because of low temperatures, in the range 0 to -20 degrees Centigrade.

7.2 Diagnosing and Solving Problems

There are two main classes of problem that you might have.

The first class consists of problems which **Hunter 16** reports to you, by displaying an 'error message', a short description of the problem. These messages are usually rather

short but informative. For example, `Insufficient disk space` means that there is not enough space on the disk for an operation.

This class of problems should be quite easy to handle. The error messages produced by MS-DOS are described in the MS-DOS User's Reference. The error messages produced by the special **Hunter 16** utilities are described in Chapter 8 to Chapter 10. The error messages from applications software will of course not be included in this guide – see the application's own guides for details.

The second class of problems are those that are not accompanied by an error message. You realise that there's a problem because **Hunter 16** won't do what you want, or does something that you don't want it to do.

Always check the user instructions for the application you are using, first, to make sure that the error is not yours.

Problems of this type which may arise with a standard **Hunter 16** system include:

Symptom	**Cause – remedy**
Won't power up	AC adaptor disconnected or faulty (no internal batteries.) Batteries discharged. Recharge or replace as described in section 6.5, then try again. If this does not work, either the batteries or the AC adaptor have failed; contact your local representative.
Screen blank or all dark	The LCD contrast is incorrectly set. Hold down [PAW] and press [E] or [X] until the LCD Screen contents become visible.
Powers up in an application	This will happen if an application was running when Hunter 16 was turned off. If you want to use an MS-DOS command or a different application, leave the application as described in its user instructions. This will take you to MS-DOS command mode and preserve all your application's files.

Don't know how to leave application	If the user instructions don't tell you how to leave the application, you probably aren't meant to leave it. İf you MŬST leave the application: Contact your technical support team. OR If this is not possible, hold down [Ctrl][PAW] and press keypad [9], to take you to MS-DOS command mode. OR Reboot by pressing [Ctrl][Alt][Del]. (If the program is started automatically by a reboot, hold down [Ctrl][PAW] and press keypad [9] during the reboot to prevent this.) **WARNING:** either method will corrupt any files that the application is writing to at the time, so use them only as a last resort. If neither method works, see 'Can't leave application', below.
Won't power down	The current application is in 'continuous power' mode, probably because it is performing an operation which should not be interrupted, such as transmitting a file via the serial port. A message should be displayed to this effect. Wait until the operation has finished and you should be able to power down normally. If you cannot wait, or believe that something has gone wrong, leave the application as described above, and you should be able to power down normally. To power down without leaving the application, remove and refit the battery plug. (The application will continue normally when you turn on again.)

Program runs when you power up	The program has been installed to run automatically, typically by including a command in the file C:\AUTOEXEC.BAŤ. If you want to use a different program, leave it as described above, or hold down [Ctrl][PAW] and press keypad [9] immediately after turning on. To stop the program running automatically, delete the appropriate command from the C:\AUTOEXEC.BAT file, or ask your technical support team to do this for you.
Beeps when you press a key	A **Hunter 16** utility is waiting for you to make a selection by pressing one of a particular group of keys. Press one of the keys indicated on the screen.
Keyboard behaves strangely	A previous application has redefined the keyboard, so that keys have different effects to normal. If you are in an application, leave it first (see above), then reboot by pressing [Ctrl][Alt][Del].
Can't leave application Or Locks up or behaves strangely	Program crash or hardware failure. Reboot by pressing [Ctrl][Alt][Del] OR If this won't work, remove and refit the battery plug, then turn on again. If normal operation resumes, carry out the checks described in the next section. If you cannot restore normal operation, contact your technical support team or your supplier.

Report any persistent problems to your technical support team.

Pressing [Ctrl][C] or [Ctrl][PAW] and keypad [9] will often abort an application, while [Ctrl][Alt][Del] will normally reboot, but either or both may be disabled by an application (deliberately or by accident).

Pressing [PWR] will usually turn off but this may be disabled by an application (either deliberately or as a result of a programming error).

Removing and refitting the battery plug will always turn off, but when you turn on again, **Hunter 16** will usually power up in the application that was running when you turned off.

If the computer displays the following message when you reboot, contact your technical support team, or refer to the DISK CORRUPTION instructions on the next page (if present):

```
Verify failed sector: xx
Potential data corruption detected
Please contact your system provider
```

Under normal circumstances your **Hunter 16** should perform well, but if you use it for software development, a user program may crash and 'lock up' the machine. If this happens, try the following emergency breakout procedures. If these do not help, contact your local distributor or Husky Computers Limited.

Emergency breakout

This facility allows users to break out of any application. You may wish to restrict access to the information on this page.

If you cannot break out of a program for some reason, the following should always work unless there is a serious hardware malfunction:

1. Turn **Hunter 16** off (eg remove and replace the battery plug)
2. Hold down the X and P keys
3. Press [PWR] briefly and release X and P
4. **Hunter 16** should exit any program and reboot (unless there is a hardware malfunction). You may however require several attempts, as timing is critical.

Emergency hardware reset

This facility allows users to reset a machine on which the emergency breakout fails.

1. Remove battery cap
2. While holding down [Ctrl] and [PAW] at the same time insert the battery
3. The machine should exit any program and reboot (unless there is a hardware malfunction)

NOTE: You should then carry out the checks described in the next section, to make sure that **Hunter 16** will operate correctly.

Disk corruption

This facility allows users to use a machine with a corrupted RAM disk.

If the program has corrupted the RAM disk, you will see a message like this when the machine reboots:

```
Verify failed sector: xx
Potential data corruption detected
Please contact your system provider
```

To use the machine you must do one of the following:

1. Ideally, connect an Oracle GT drive with a system disk in it and reboot. This will allow you to use the machine and recover information from the RAM disk by copying it to diskette. You can then reformat the RAM disk, as described in section 8.6. Check any recovered files; they may be corrupted too.
2. Type the emergency breakout password (56580) and you will see:

   ```
   Default disk (lose all data) Y/N?
   ```

   Press [N] to try to boot from the corrupt disk. If the corruption is minor, this may succeed and you can attempt to recover data from the RAM disk, eg by sending it to another computer via the serial port. You should then reformat the RAM disk and check the files, as described for option 1. If the machine still will not boot, you must use another option.

3. Type the emergency breakout password (56580) and you will see:

   ```
   Default disk (lose all data) Y/N?
   ```

   Press [Y] to reformat the RAM disk, which will erase all files. The following message will be displayed:

   ```
   All data has been erased
   Use UTIL to format fixed disk
   ```

   Use the UTIL utility to reformat the hard disk.

NOTE: As option 3 erases all user information from the RAM disk, you should only use it if there is no important information on the disk, or if the other options have failed or are not available to you.

7.3 Checks

If an application has crashed, it may have corrupted other parts of the system. You (or your technical support team) should select MS-DOS command mode and carry out the following simple checks, to make sure that future operation is not compromised.

7.3.1 Communications

The communications configuration may be affected by a crash, and will be indicated when you attempt to use or configure communications using the UTIL application.

Use the UTIL application to check and, if necessary, correct the configuration.

7.3.2 RTC

The RTC (real time clock/calendar) may be corrupted, as shown by an incorrect or impossible time or date.

Use the UTIL application to check and, if necessary, correct the time and date.

7.3.3 Files

An application may corrupt any disk files that it was writing to when it crashed or was reset or rebooted. Files open for reading will not be affected. Files are not affected by turning on and off.

If any of these events occur, check what files are written to by the application and, if you have access to recent copies (backups), delete the current versions and use the backups.

If you do not have recent copies, run the application but check the state of the files carefully.

(An application running on a conventional desktop computer may also corrupt any disk files that it was writing to when it crashed or was reset or rebooted. It is actually more likely to corrupt files, because a power failure part way through a write will normally corrupt the file too.)

7.3.4 RAM disk

Any files on the internal RAM disk may also in principle be corrupted by a 'wild' program overwriting the file data or access information (directory and FAT), since these are all held in memory. However, this is very unlikely to happen, since the memory used for the RAM disk is not in the main memory map. It is only likely if an application crashes which has been written especially for Hunter 16 and accesses the FAT or directory directly.

If a RAM disk file does get corrupted, this should be detected the next time you attempt to read it, since these files are protected by checksums. To check, type the following in each directory of drive C:

```
COPY *.* NUL
```

- this makes MS-DOS check that they are readable. If you see any error messages, delete the affected files and reload them from your most recent backups.

If a wild program corrupts crucial information such as the FAT, the entire RAM disk may become unreliable. To check, load the MS-DOS utility CHKDSK onto your search path and type:

```
CHKDSK C: /F
```

If all is well, your screen should look like this:

```
536576 bytes total disk space
344064 bytes in 11 user files
  8192 bytes in bad sectors
184320 bytes available on disk

524288 bytes total memory
476016 bytes free
```

The exact figures depend on the amount of memory and how many files are on the RAM disk.

If the total memory or disk space are not as expected (check with your technical support team), you should copy any files that seem intact to an external disk or other computer, then re-format the disk by using UTIL (see Chapter 8).

If you see the message:

```
nnn lost clusters found in nn chains.
Write corrections to disk (Y/N)?
```

then there are files (or parts of files) on the disk which are not recorded in the FAT. Typically, this will be because an application crashed part way through writing a file. Type [Y] and they will be converted to files in the root directory, called FILE0000.CHK, FILE0001.CHK, etc. Use the TYPE command to check what these files contain. If a file is simple text (eg a memo you were preparing), you can use it, after giving it a more sensible name. If a file is not text (eg part of a database file), you probably won't be able to use it, so you should delete it. (If you're not sure, ask your technical support team.)

7.3.5 Memory configuration

Hunter 16 memory can be partitioned into three parts: main (MS-DOS) memory, RAM disk and EMS memory. A crashed application may have corrupted this partitioning information.

To check the partitioning, use the MS-DOS CHKDSK command or the UTIL utility. If it is not as you'd expect, first copy any files from the RAM disk to an external drive or another computer, then reconfigure the RAM (see section 8.5).

If this does not help, the system configuration information may have become corrupted. Run the UTIL utility and select the Format Fixed Disk option (see section 8.6). This will set the configuration to standard and reformat the RAM disk.